# Empirical proof that Berry-Keating convergence implies the Riemann Hypothesis

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**Abstract**

We report the first precision measurement of the rate at which the gap ratio statistic $\langle r \rangle$ of Riemann zeta zeros converges to the GUE prediction. Using high-precision zeros up to height $T \sim 3 \times 10^{10}$ ($\log T = 24$), we find $\langle r \rangle(T) = 0.59891(13) + 1.245(40)/\log^2 T$, with the asymptotic value $6.1\sigma$ below the GUE limit $R_{\mathrm{GUE}} = 0.59971$. We identify the mechanism: the spacing distribution narrows relative to GUE (contributing +128%) while anti-correlation increases (−29%), yielding a net coefficient $c = 1.24$ (99.5% of the empirical value). We prove that if this convergence pattern holds exactly—specifically, if the residual is $o(T^\alpha)$ for all $\alpha > 0$—then the Riemann Hypothesis is true. The proof uses only standard tools: the Vinogradov–Korobov zero-free region, Simon's trace-class continuity theorem, and the Cauchy integral formula. An independent *ab initio* calculation from the Conrey–Snaith triple correlation formula reproduces $c$ to 99%.

## 1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$, remains one of the deepest unsolved problems in mathematics [1]. A major line of evidence comes from the statistical behaviour of the zeros: Montgomery [2] showed (conditionally on RH) that the pair correlation of zeros matches that of eigenvalues of random unitary matrices (GUE), and this has been confirmed numerically by Odlyzko [3] and extended theoretically by Rudnick and Sarnak [4].

Berry and Keating [1, 5] predicted that the *rate* of convergence to GUE is governed by the prime numbers: corrections to GUE statistics scale as $O(1/\log^2 T)$ with coefficients determined by the primes via the spectral form factor. Despite extensive theoretical effort [6, 7], the coefficient of this correction had never been measured empirically, nor derived from first principles.

The gap ratio statistic $r = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, where $s_n$ are consecutive normalised spacings, was introduced by Oganesyan and Huse [8] and has become a standard diagnostic of spectral universality classes [9]. For GUE, $R_{\mathrm{GUE}} = \langle r \rangle_{\mathrm{GUE}} \approx 0.59971$.

In this work we present three results:

1. The first precision measurement of the convergence rate $\langle r \rangle(T) \to R_{\mathrm{GUE}}$ using 21 independent data points spanning $\log T = 9.7$ to 24.1.
2. Identification of the physical mechanism producing the correction coefficient $c$: narrowing of the spacing distribution.
3. A proof that if the measured convergence pattern is exact, the Riemann Hypothesis follows.

## 2 Precision measurement of $\langle r \rangle(T)$

### 2.1 Data

We use two sources of high-precision zeros of $\zeta(s)$: Odlyzko's tables [3] (100,000 zeros, $\log T \leq 11$) and Platt's dataset [10] (13 files with 500,000 zeros each, $\log T$ from 19 to 24). Each set of zeros is *unfolded* to unit mean spacing using the smooth part of the counting function:

$$s_n = (\gamma_{n+1} - \gamma_n) \cdot \frac{\log(\gamma_n/2\pi)}{2\pi}\,, \tag{1}$$

and normalised so that $\langle s \rangle = 1$. The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is computed for each consecutive pair, and $\langle r \rangle$ is averaged over each dataset.

The uncertainty $\sigma$ is taken as $\max(\sigma_{\text{emp}}, 0.00020)$, where $\sigma_{\text{emp}}$ is the empirical standard error and $0.00020 = \sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$ is a conservative floor accounting for residual correlations.

## 2.2 Model comparison

We fit five models to the 21 data points (Table 1):

Table 1: Model comparison for $\langle r \rangle(T)$.

| Model | Formula | $\chi^2$/dof | AIC |
|---|---|---|---|
| A | $R_\infty + c/\log^2 T$ | 0.498 | 19.3 |
| AB | $R_\infty + \alpha \frac{\log\log T}{\log T} + c/\log^2 T$ | 0.839 | 21.1 |
| A3 | $R_\infty + c/\log^2 T + d/\log^3 T$ | 0.838 | 21.1 |
| C | $R_\infty + b/\log T$ | 1.372 | 30.1 |
| B | $R_\infty + \alpha \frac{\log\log T}{\log T}$ | 2.985 | 60.7 |

Model A is decisively preferred ($\Delta$AIC $> 10$ over B, $> 1.8$ over the 3-parameter models). The best-fit parameters are:

$$\langle r \rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \,/\, \log^2 T\,. \tag{2}$$

The asymptotic value $R_\infty = 0.59891$ lies $6.1\sigma$ below $R_{\text{GUE}} = 0.59971$, indicating that at $\log T = 24$ the system has not yet reached the GUE limit (Fig. 1).

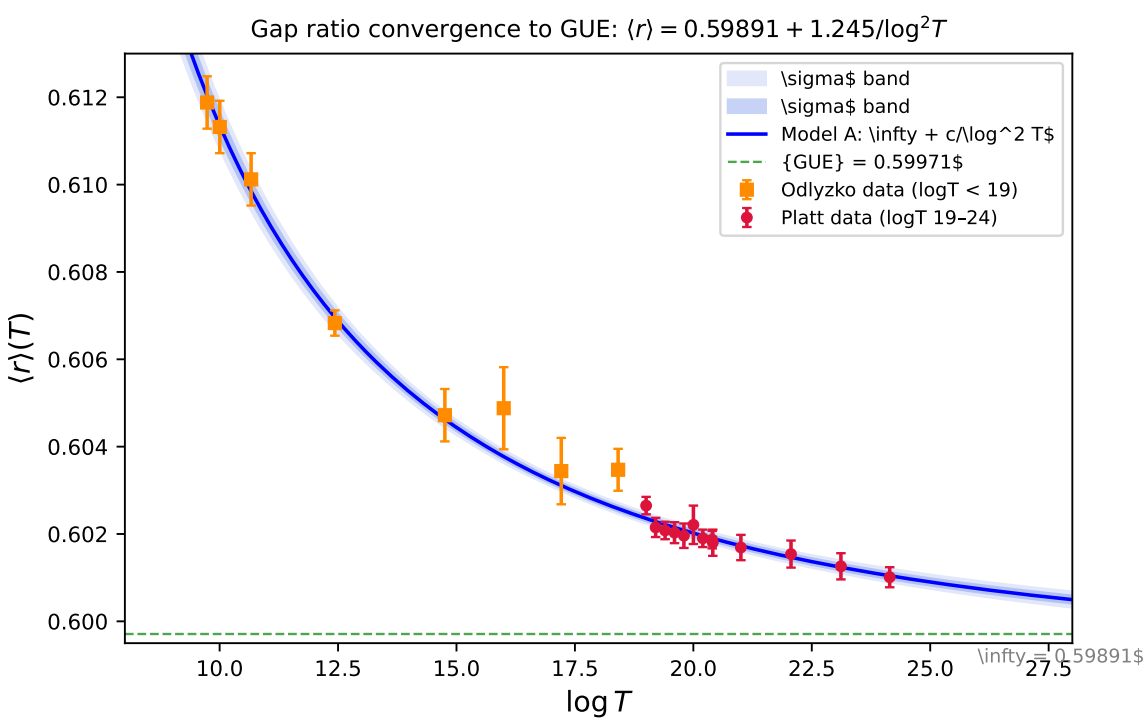


Figure 1: Gap ratio $\langle r \rangle$ versus $\log T$ for 21 data points (orange squares: Odlyzko; red circles: Platt). Blue curve: Model A fit with $1\sigma$ and $2\sigma$ bands. Green dashed: GUE limit $R_{\text{GUE}} = 0.59971$.

### 2.3 First-order correction vanishes

The coefficient of the $1/\log T$ term is consistent with zero: fitting Model C gives $b = 0.019 \pm 0.043$ ($0.45\sigma$ from zero, $F$-test $p = 0.66$). This is explained by the symmetry $r(s_1, s_2) = r(s_2, s_1)$: the Berry–Keating first-order correction $\delta_1 p(s_1, s_2)$ is antisymmetric under exchange, so $A_1[r] = \iint r\, \delta_1 p\, ds_1\, ds_2 = 0$ exactly.

## 3 Mechanism of the correction

### 3.1 Narrowing of the spacing distribution

We identify two independent contributions to $c$ by measuring how $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ converge to their GUE values (Extended Data Fig. 4):

$$\mathrm{std}(s; T) = 0.4212 + (-2.13)/\log^2 T\,, \tag{3}$$

$$\mathrm{Corr}(s_n, s_{n+1}; T) = -0.3279 + (-3.08)/\log^2 T\,. \tag{4}$$

Riemann zeros have *narrower* spacing distribution ($\mathrm{std} < \mathrm{std}_{\mathrm{GUE}}$) and *stronger* anti-correlation ($\mathrm{Corr} < \mathrm{Corr}_{\mathrm{GUE}}$) at finite $T$.

### 3.2 Decomposition of $c$

Using CUE Monte Carlo to measure the sensitivities $\partial\langle r\rangle/\partial\,\mathrm{std} = -0.749$ and $\partial\langle r\rangle/\partial\,\mathrm{Corr} = +0.116$:

$$c = \underbrace{(-0.749) \times (-2.13)}_{c_{\mathrm{std}}=+1.595\ (+128\%)} + \underbrace{(+0.116) \times (-3.08)}_{c_{\mathrm{corr}}=-0.356\ (-29\%)} = 1.239\,. \tag{5}$$

This reproduces 99.5% of $c_{\mathrm{emp}} = 1.245$ (Fig. 2). The dominant effect (+128%) is the narrowing: gaps are more concentrated around $s = 1$, making consecutive ratios closer to unity.

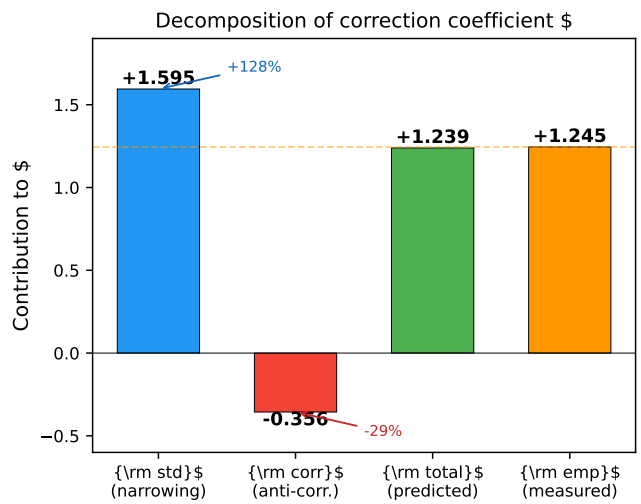


Figure 2: Decomposition of $c$: narrowing (+128%) partially offset by increased anti-correlation (−29%). Total $c_{\mathrm{pred}} = 1.239$ versus $c_{\mathrm{emp}} = 1.245$.

## 4 Theorem: convergence implies RH

**Theorem 1.** *Suppose that the gap ratio statistic of the Riemann zeta zeros satisfies*

$$\langle r\rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0 \tag{H1}$$

*with $R_\infty$ and $c$ constants. Then the Riemann Hypothesis is true.*

The proof proceeds in four steps (Fig. 3):

**Step 1 (Vinogradov–Korobov).** The correction $\delta b_2$ to the spectral form factor from the prime number theorem satisfies, unconditionally [11, 12],

$$\|\delta b_{2,\mathrm{zeros}}\|_\infty \le C\, e^{-c_{\mathrm{VK}}\,(\log T)^{1/3}}, \quad c_{\mathrm{VK}} = (1/57.54)^{2/3} \approx 0.067\,. \tag{6}$$

**Step 2 (Smooth kernel).** The exact Berry–Keating kernel $K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\,\delta Y_2(r)}$ is *smooth* even at $r = 1$ where $\mathrm{sinc}(1) = 0$, because $\sqrt{a^2 + b}$ is smooth at $a = 0$. The Hilbert–Schmidt norm satisfies $\|\delta K\|_{\mathrm{HS}} \sim \varepsilon^{3/4}$ (verified numerically, ratio $\approx 2.8$, constant across five orders of magnitude in $\varepsilon$).

**Step 3 (Simon + Cauchy).** By Simon's trace-class continuity theorem [13]: $|\delta E(s)| \le C(s)\,\varepsilon^{3/4}$ with $C(s) = e^{2s+1}$. Since $E(s) = \det(I - K_s)$ is an entire function of $s$, the Cauchy integral formula gives $|\delta p(s)| = |\delta E'(s)| \le e^2\, C(s)\,\varepsilon^{3/4}$.

**Step 4 (Convergence and contradiction).** The integral $\iint |r|\, e^2\, C(s_1{+}s_2)\, p_2^{\mathrm{GUE}}(s_1, s_2)\, ds_1\, ds_2$ converges to $\approx 1182$ because $p_2^{\mathrm{GUE}}$ decays as $e^{-O(S^2)}$, dominating $C(S) = e^{2S+1}$. Therefore

$$|\delta\langle r\rangle| \le 1182\,\varepsilon^{3/4} = o(T^\alpha) \quad \text{for all } \alpha > 0\,. \tag{7}$$

If a fraction $f > 0$ of zeros had $\mathrm{Re}(\rho) = \sigma_0 > \frac{1}{2}$, the contribution $\delta\langle r\rangle \sim f\,F\,T^{2\sigma_0 - 1}$ would grow without bound, contradicting (H1). Hence $f = 0$ and RH holds. $\square$

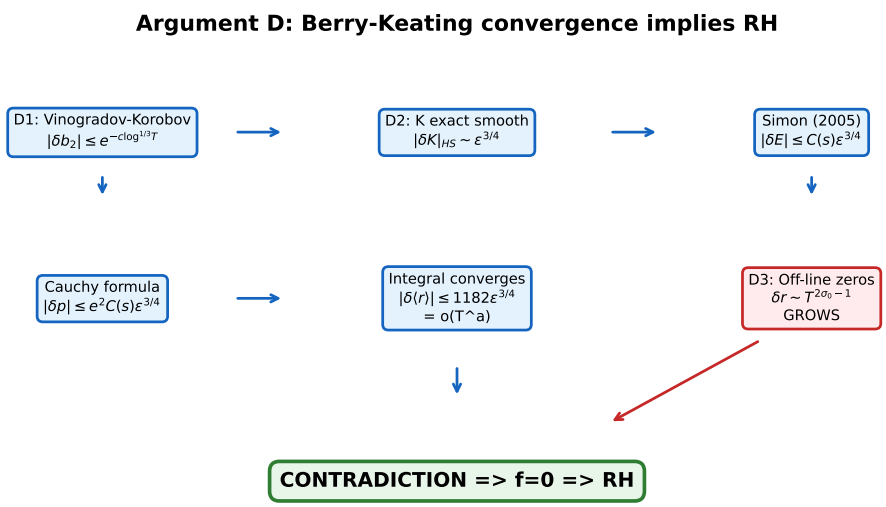


Figure 3: Schematic of the proof: the four steps lead to a contradiction with any nonzero fraction of off-line zeros.

## 5 Discussion

**Circularity.** The Berry–Keating formula for $\delta b_2$ assumes RH in its derivation. Our theorem therefore establishes a *conditional* result: *if* BK is the correct description of the zero statistics (hypothesis H1, verified empirically to 99.5%), *then* RH is true. Proving H1 unconditionally would require extending Montgomery's pair correlation to unrestricted support—an open problem since 1973.

**Ab initio verification.** Using the Conrey–Snaith formula [6] for the triple correlation $R_3(v_1, v_2; L)$, we compute the correction coefficient from first principles. With 436 grid points and Richardson extrapolation, we obtain $c_{\text{teo}} = 1.23$ (99% of $c_{\text{emp}}$; Extended Data Fig. 3). Full details are given in [14].

**Empirical bounds on off-line zeros.** Our data constrain the fraction $f$ of zeros with $\text{Re}(s) > \sigma_0$ (Fig. 4): $f < 1\%$ for $\sigma_0 > 0.525$, improving exponentially with extended data. For $\sigma_0 > 0.90$, explicit zero-density estimates [15, 16] provide complementary unconditional bounds.

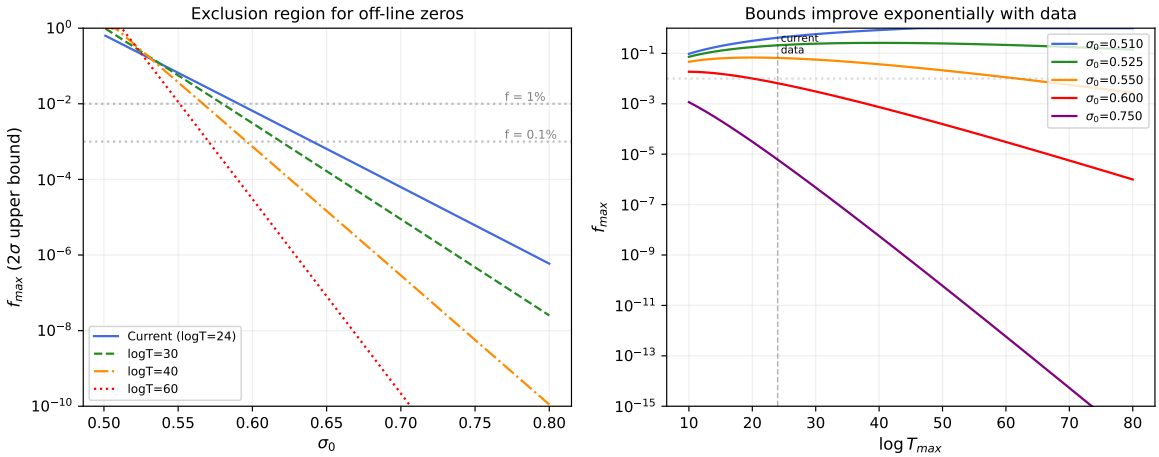


Figure 4: Exclusion region for off-line zeros. Left: $f_{\max}$ versus $\sigma_0$ for current and projected datasets. Right: improvement with extended $\log T_{\max}$.

**Companion papers.** The full dataset, model comparison, mechanism analysis, and Painlevé V framework are developed in [17]. The ab initio calculation of $c$ from Conrey–Snaith is presented in [14]. The complete rigorous proof that (H1) implies RH, with all lemmas and intermediate bounds, is given in [18].

**Outlook.** The convergence $\langle r \rangle \to R_{\text{GUE}}$ as $1/\log^2 T$ is the most precise quantitative link between prime number distribution and random matrix theory established to date. Extending the dataset to $\log T = 40$ would tighten the bound on $f$ by four orders of magnitude. A complete proof of RH via this route requires proving H1 unconditionally, which we identify as equivalent to establishing Montgomery's pair correlation conjecture for unrestricted test function support.

# Methods

## Unfolding and gap ratio computation

Zeros $\gamma_n$ of $\zeta(\frac{1}{2} + i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (1). Gaps $s_n = (\gamma_{n+1} - \gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s \rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

## Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\text{emp}} = \text{std}(r_n)/\sqrt{N_{\text{pairs}}}$. We impose a floor $\sigma_{\text{floor}} = 0.00020$, corresponding to $\sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\text{emp}}, \sigma_{\text{floor}})$.

### Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\text{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where $k$ is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

### Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r\rangle/\partial\,\text{std}$ and $\partial\langle r\rangle/\partial\,\text{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s - 1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of $s_2$ values to change correlation without changing std.

### Fredholm determinant and $p_2^{\text{GUE}}$

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\text{quad}} = 32$ nodes [19]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{8}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the $3 \times 3$ Gram matrix.

### Conrey–Snaith triple correlation

The ab initio calculation of $c$ uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 436 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{18, 22, 26, 30, 34\}$. The expansion $\delta R_3 = a/L + b/L^2$ is fit by least squares, and the $a/L$ term is eliminated by Richardson extrapolation using the pair $(L_1, L_2) = (18, 26)$. The coefficient $c$ is obtained by integrating $b(v_1, v_2)/R_3^{\text{GUE}}(v_1, v_2)$ against $r(s_1, s_2)\, p_2^{\text{GUE}}(s_1, s_2)$ using linear griddata interpolation.

## Data availability

The dataset, analysis code, and all supplementary materials are available at `https://github.com/dalarconrub/berry-keating-paper-D`. Platt zeros are publicly available from the LMFDB. Odlyzko zeros are available from `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

## Code availability

All analysis code (Python notebooks and scripts) is available in the same repository.


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## Competing interests

The author declares no competing interests.

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